Lecture Notes

BAYESIAN ANALYSIS

DSA 8505

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Contents

5 Bayesian Inference in Regression Models

Bayesian inference provides a probabilistic approach to estimating regression parameters by incorporating prior knowledge. The posterior distribution is obtained using Bayes' theorem:

$$P(\theta|D) \propto P(D|\theta)P(\theta)$$

where:

- $P(\theta|D)$ is the posterior distribution of parameters given data D,
- $P(D|\theta)$ is the likelihood function,
- $P(\theta)$ is the prior distribution of parameters.

5.1 Bayesian Inference for Binomial Regression

In binomial regression, the response variable follows a binomial distribution:

$$Y_i \sim \text{Binomial}(n_i, p_i)$$

where Y_i represents the number of successes out of n_i trials, and p_i is the probability of success.

A common link function used in binomial regression is the logit function:

$$\log\left(\frac{p_i}{1-p_i}\right) = X_i\beta$$

which ensures that p_i remains between 0 and 1.

A Bayesian approach requires specifying:

- **Likelihood**: Based on the binomial distribution:

$$P(Y_i|\beta) = \binom{n_i}{Y_i} p_i^{Y_i}(1-p_i)^{n_i-Y_i}$$

- **Priors**: Common choices include normal priors for β:

$$\beta \sim \mathcal{N}(0, \sigma^2)$$

where σ^2 is a variance hyperparameter that can be set based on prior knowledge.

- **Posterior Distribution**: Using Bayes' theorem, we update our beliefs about β given the data:

  $$P(\beta|Y) \propto P(Y|\beta)P(\beta)$$

  This posterior is typically estimated using Markov Chain Monte Carlo (MCMC) methods.

5.1.1 Worked Example: Bayesian Logistic Regression

Suppose we are modeling the probability of a student passing an exam ($Y_i = 1$) based on the number of study hours (X_i). We assume a logistic regression model:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_i$$

Given the following dataset:

Hours Studied (X_i)	Passed Exam (Y_i)
1	0
2	0
3	1
4	1
5	1

We assume the following priors:

$$\beta_0 \sim \mathcal{N}(0, 10)$$

$$\beta_1 \sim \mathcal{N}(0, 10)$$

The likelihood function is:

$$P(Y|\beta) = \prod_{i=1}^{n} p_i^{Y_i}(1-p_i)^{1-Y_i}$$

Using MCMC methods (such as Gibbs sampling or Hamiltonian Monte Carlo), we estimate the posterior distributions of β_0 and β_1. The resulting posterior means provide an estimate of the effect of study hours on passing the exam. See the results HERE

After running MCMC, we obtain posterior samples for β_0 and β_1. Suppose the estimated values are:

$$\beta_0 = -11, \quad \beta_1 = 4.9$$

This means that for each additional hour of study, the log-odds of passing increases by 0.8. Converting back to probability:

$$p = \frac{1}{1 + e^{-(-1.5+0.8X_i)}}$$

For a student studying 3 hours:

$$p = \frac{1}{1 + e^{-(-1.5+0.8\times 3)}} \approx 0.73$$

Thus, the probability of passing after studying for 3 hours is approximately 73%.

Ordinal regression is used when the response variable consists of ordered categories with unknown spacing between levels. Typical examples include Likert-scale survey responses, disease severity classifications, and satisfaction ratings. Classical linear regression is inappropriate for such data because it assumes equal distances between outcome categories.

Bayesian ordinal regression provides a principled framework for modeling ordered outcomes while incorporating prior information and quantifying uncertainty through posterior distributions.

5.2 Bayesian Inference for ordinal regression

Let Y_i be an ordinal response variable with K ordered categories $\{1, 2, \ldots, K\}$. The proportional odds model is defined through cumulative probabilities:

$$P(Y_i \leq k) = \frac{\exp(\alpha_k - X_i\beta)}{1 + \exp(\alpha_k - X_i\beta)}, \quad k = 1, \ldots, K-1$$

where:

- X_i is a vector of predictors,
- β is a vector of regression coefficients,
- α_k are cutpoints (thresholds) separating adjacent categories.

The model assumes **proportional odds**, meaning that the effect of predictors β is constant across all category boundaries.

5.2.1 Latent Variable Interpretation

The proportional odds model can be derived from an unobserved latent variable Z_i:

$$Z_i = X_i\beta + \varepsilon_i, \quad \varepsilon_i \sim \text{Logistic}(0, 1)$$

The observed ordinal outcome is obtained by thresholding Z_i:

$$Y_i = \begin{cases} 1 & Z_i \leq \alpha_1 \\ 2 & \alpha_1 < Z_i \leq \alpha_2 \\ \vdots & \\ K & Z_i > \alpha_{K-1} \end{cases}$$

This formulation leads directly to the cumulative logit model.

5.2.2 Bayesian Inference Components

Bayesian ordinal regression consists of the following components:

Priors

- Regression coefficients:
$$\beta \sim \mathcal{N}(0, 10^2)$$
- Cutpoints (ordered):
$$\alpha_k \sim \mathcal{N}(0, 5^2), \quad \alpha_1 < \alpha_2 < \cdots < \alpha_{K-1}$$

Likelihood

The probability of observing category k is given by:

$$P(Y_i = k) = \text{logit}^{-1}(\alpha_k - X_i\beta) - \text{logit}^{-1}(\alpha_{k-1} - X_i\beta)$$

Posterior Inference

Posterior distributions are obtained using Markov Chain Monte Carlo (MCMC) methods such as Hamiltonian Monte Carlo (HMC).

Worked Example

Question

A university conducts a course evaluation where students rate a course using three ordered categories:

$$1 = \text{Poor}, \quad 2 = \text{Average}, \quad 3 = \text{Good}$$

The evaluation is modeled using a **proportional odds (ordinal logistic) regression model**. The cumulative probabilities are defined as:

$$P(Y \leq k \mid X) = \frac{\exp(\alpha_k - X\beta)}{1 + \exp(\alpha_k - X\beta)}, \quad k = 1, 2$$

where:

- Y denotes the ordinal course rating,
- X represents the number of hours a student studies per week,
- β is the regression coefficient,
- α_k are the cutpoints separating the rating categories.

Assume the following parameter values:

$$\alpha_1 = -1, \quad \alpha_2 = 1, \quad \beta = 0.5$$

A student studies for $X = 4$ hours per week.

1. Compute the linear predictor η.
2. Using the proportional odds model, calculate:

   (a) $P(Y \leq 1)$,
   (b) $P(Y \leq 2)$.

3. Derive the probabilities:
   (a) $P(Y = 1)$,
   (b) $P(Y = 2)$,
   (c) $P(Y = 3)$.
4. Verify that the category probabilities sum to one.
5. Interpret the results in the context of course evaluation.

Step 1: Proportional Odds Model

The proportional odds (cumulative logit) model is defined as:

$$P(Y \leq k \mid X) = \frac{\exp(\alpha_k - X\beta)}{1 + \exp(\alpha_k - X\beta)}, \quad k = 1, 2$$

where:

- Y is the ordinal response variable,
- X is the predictor variable,
- β is the regression coefficient,
- α_k are the cutpoints.

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Step 2: Given Parameter Values

Assume the following values:

$$\alpha_1 = -1, \quad \alpha_2 = 1, \quad \beta = 0.5$$

For a student studying:

$$X = 4 \text{ hours per week}$$

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Step 3: Compute the Linear Predictor

The linear predictor is given by:

$$\eta = X\beta$$

Substituting the given values:

$$\eta = 4 \times 0.5 = 2$$

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Step 4: Compute the Cumulative Probability $P(Y \leq 1)$

Using the proportional odds formula:

$$P(Y \leq 1) = \frac{\exp(\alpha_1 - \eta)}{1 + \exp(\alpha_1 - \eta)}$$

Substitute $\alpha_1 = -1$ and $\eta = 2$:

$$P(Y \leq 1) = \frac{\exp(-1 - 2)}{1 + \exp(-1 - 2)} = \frac{\exp(-3)}{1 + \exp(-3)}$$

Since:

$$\exp(-3) \approx 0.0498$$

we obtain:

$$P(Y \leq 1) = \frac{0.0498}{1 + 0.0498} \approx 0.047$$

—

Step 5: Compute the Cumulative Probability $P(Y \leq 2)$

$$P(Y \leq 2) = \frac{\exp(\alpha_2 - \eta)}{1 + \exp(\alpha_2 - \eta)}$$

Substitute $\alpha_2 = 1$ and $\eta = 2$:

$$P(Y \leq 2) = \frac{\exp(1 - 2)}{1 + \exp(1 - 2)} = \frac{\exp(-1)}{1 + \exp(-1)}$$

Since:

$$\exp(-1) \approx 0.3679$$

we obtain:

$$P(Y \leq 2) = \frac{0.3679}{1 + 0.3679} \approx 0.269$$

—

Step 6: Compute Category Probabilities

The probability of each category is obtained from the cumulative probabilities.

Category 1 (Poor)

$$P(Y = 1) = P(Y \leq 1) = 0.047$$

Category 2 (Average)

$$P(Y = 2) = P(Y \leq 2) - P(Y \leq 1)$$

$$P(Y = 2) = 0.269 - 0.047 = 0.222$$

Category 3 (Good)

$$P(Y = 3) = 1 - P(Y \leq 2)$$

$$P(Y = 3) = 1 - 0.269 = 0.731$$

—

Step 7: Probability Check

To verify that the probabilities sum to one:

$$P(Y = 1) + P(Y = 2) + P(Y = 3) = 0.047 + 0.222 + 0.731 = 1.000$$

—

Final Interpretation

For a student studying 4 hours per week:

- Probability of a **Poor** rating is 4.7%,
- Probability of an **Average** rating is 22.2%,
- Probability of a **Good** rating is 73.1%.

The positive value of $\beta = 0.5$ indicates that increased study time shifts probability mass toward higher ordinal categories.

5.3 Simulated Dataset

A synthetic dataset is generated where:

- Predictor: weekly study hours
- Response: course rating (Poor, Average, Good)

The latent variable follows:

$$Z_i = 0.6X_i + \varepsilon_i, \quad \varepsilon_i \sim \text{Logistic}(0, 1)$$

5.4 Bayesian Ordinal Regression in Python

5.4.1 Data Generation

```
import numpy as np
import pymc as pm
import arviz as az

np.random.seed(42)

n = 120
x = np.random.uniform(0, 10, size=n)

beta_true = 0.6
cutpoints_true = [-1.0, 1.2]

z = beta_true * x + np.random.logistic(size=n)
y = np.digitize(z, cutpoints_true)
```

5.4.2 Model Specification

```
x_std = (x - x.mean()) / x.std()

with pm.Model() as model:

    beta = pm.Normal("beta", mu=0, sigma=10)

    cutpoints = pm.Normal(
        "cutpoints",
        mu=0,
        sigma=5,
        shape=2,
        transform=pm.distributions.transforms.ordered
    )

    eta = beta * x_std

    y_obs = pm.OrderedLogistic(
        "y_obs",
        eta=eta,
        cutpoints=cutpoints,
        observed=y
    )

    trace = pm.sample(
        2000,
        tune=2000,
        target_accept=0.95,
        return_inferencedata=True
    )
```

5.5 Model Diagnostics

Convergence is assessed using:

- Trace plots
- Effective sample size (ESS)
- $\hat{R}$ statistics

```
az.summary(trace)
az.plot_trace(trace)
```

5.6 Posterior Interpretation

- A positive posterior mean of β indicates that increased study hours raise the probability of higher ratings.
- Cutpoints define the difficulty of transitioning between ordinal categories.
- Credible intervals quantify parameter uncertainty.

5.7 Applications

Bayesian ordinal regression is widely used in:

- Medicine: disease severity classification
- Social sciences: Likert-scale survey analysis
- Economics: consumer satisfaction modeling
- Education: course evaluation studies

5.8 Conclusion

Bayesian ordinal regression provides a flexible and interpretable framework for modeling ordered categorical outcomes. By combining the proportional odds assumption with prior information and MCMC-based inference, it enables robust uncertainty quantification and principled decision-making in a wide range of applied settings.

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6 Bayesian Hierarchical Models

6.1 Understanding Bayesian Hierarchical Models

Bayesian hierarchical models (BHMs) are statistical models that involve multiple levels of parameters, allowing for structured dependency among observations. These models are particularly useful when dealing with grouped data, where each group has its own parameters that are linked through a common higher-level distribution.

6.1.1 Definition and Structure

- **Likelihood Function** $P(y|\theta)$
  - Defines the probability distribution of the observed data y given the model parameters θ.
  - Example: If we assume normally distributed data:

$$y_i \sim \mathcal{N}(\mu, \sigma^2)$$

- **Prior Distributions** $P(\theta)$
  - Expresses prior beliefs about the parameters θ before observing data.
  - Example: A normal prior on μ:

$$\mu \sim \mathcal{N}(\mu_0, \tau^2)$$

- **Hyperpriors** $P(\lambda)$
  - Represents higher-level uncertainty by placing distributions on prior parameters.
  - Example: An inverse-gamma prior for σ^2:

$$\sigma^2 \sim \text{Inverse-Gamma}(\alpha, \beta)$$

Example: Suppose we have test scores from different schools, where each school has its own mean but shares a common variance structure:

$$\begin{aligned}
Y_{ij}|\mu_j, \sigma^2 &\sim \mathcal{N}(\mu_j, \sigma^2), \quad j = 1, \ldots, J; \quad i = 1, \ldots, n_j \\
\mu_j|\mu_0, \tau^2 &\sim \mathcal{N}(\mu_0, \tau^2) \\
\mu_0 &\sim \mathcal{N}(0, 10^2), \quad \tau^2 \sim \text{Inverse-Gamma}(2, 1) \\
\sigma^2 &\sim \text{Inverse-Gamma}(2, 1)
\end{aligned}$$

6.1.2 Hierarchical Structure Representation

Bayesian hierarchical models are structured in three levels:

1. **Data Level (Observations)**:
   - The first level of the hierarchy represents the observed data, denoted as y, which follows a likelihood model governed by unknown parameters θ.
   - This level captures the variability in the data, typically assuming an underlying probability distribution such as Normal, Poisson, or Binomial.
   - Example: If we are modeling student test scores y_{ij} in school j, the likelihood function might be:
   $$y_{ij} \sim \mathcal{N}(\mu_j, \sigma^2)$$
   where μ_j represents the mean test score for school j, and σ^2 represents measurement variability.

2. **Parameter Level (Prior Distributions)**:
   - The second level introduces prior distributions on the parameters θ to encode domain knowledge and uncertainty before observing data.
   - These priors help stabilize estimates, especially in cases with small sample sizes or limited data.
   - Example: We may assume that the school-specific mean test scores μ_j follow a normal distribution around an overall mean:
   $$\mu_j \sim \mathcal{N}(\mu_0, \tau^2)$$
   where μ_0 is the overall population mean, and τ^2 is the variance capturing differences across schools.

3. **Hyperparameter Level (Hyperpriors)**:
   - The third level further models the uncertainty in prior distributions using hyperpriors on hyperparameters τ^2.
   - Hyperpriors introduce additional flexibility and allow the model to adapt based on the data structure.
   - Example: We can model the prior variance τ^2 with an inverse-gamma hyperprior to allow for automatic regularization:
   $$\tau^2 \sim \text{Inverse-Gamma}(\alpha, \beta)$$
   where α and β are chosen based on prior beliefs about the variability in means across different groups.

This hierarchical approach enables **borrowing of strength**, where information from all schools informs the estimation of each school's mean.

6.1.3 Posterior Predictive Checks

Posterior predictive distributions are used to check if simulated data resemble real data:

$$Y_{rep}|Y \sim p(Y_{rep}|\theta)$$

If replicated data deviate significantly from observed data, the model might be misspecified.

6.1.4 Information Criteria

Common Bayesian model selection metrics include:

- **Deviance Information Criterion (DIC)**:

  $$DIC = D(\hat{\theta}) + 2p_D,$$

  where p_D is the effective number of parameters.

- **Widely Applicable Information Criterion (WAIC)**: A fully Bayesian alternative to AIC.

6.2 Bayesian Model Checking and Model Selection

Evaluating the adequacy of a Bayesian model is crucial to ensure that it properly represents the observed data. Two commonly used techniques for model checking and selection are **Posterior Predictive Checks** and **Information Criteria**.

6.2.1 Posterior Predictive Checks

Posterior predictive checks assess the validity of a model by comparing replicated data generated from the posterior distribution to the observed data. The idea is to simulate new datasets under the fitted model and check whether these resemble the real data. If significant discrepancies are found, the model may be misspecified.

Posterior Predictive Distribution: Given observed data Y and a posterior distribution $p(\theta|Y)$, the posterior predictive distribution is defined as:

$$Y_{rep}|Y \sim p(Y_{rep}|Y) = \int p(Y_{rep}|\theta)p(\theta|Y)d\theta.$$

Here, - Y_{rep} represents new (replicated) data generated from the posterior. - θ are the model parameters. - $p(Y_{rep}|\theta)$ is the likelihood of new data given parameters. - $p(\theta|Y)$ is the posterior distribution of the parameters.

Checking for Model Fit: The replicated datasets Y_{rep} should be statistically similar to the observed dataset Y. Discrepancies may indicate model misspecification. One common way to quantify such discrepancies is through test statistics:

$$T(Y) \text{ vs. } T(Y_{rep})$$

where $T(\cdot)$ is a function measuring some characteristic of the data (e.g., mean, variance, quantiles). The posterior predictive p-value is computed as:

$$p_{ppc} = P(T(Y_{rep}) \geq T(Y)|Y).$$

A good fit is indicated when the posterior predictive p-value p_{ppc} is close to 0.5.

- If $p_{ppc} \approx 0.5$: The replicated data Y_{rep} behaves similarly to the observed data Y, meaning the model is likely well-specified.
- If p_{ppc} is close to 0 or 1: The replicated data differs significantly from the observed data, suggesting model misspecification.

A posterior predictive p-value too extreme (very small or very large) indicates that the model is not capturing key characteristics of the data well, requiring further model refinement.

6.2.2 Information Criteria

Information criteria provide a quantitative way to compare Bayesian models by balancing goodness-of-fit and model complexity. Two commonly used criteria are the **Deviance Information Criterion (DIC)** and the **Widely Applicable Information Criterion (WAIC).**

Deviance Information Criterion (DIC): DIC is a generalization of AIC for Bayesian models and is defined as:

$$DIC = D(\hat{\theta}) + 2p_D,$$

where:
- $D(\hat{\theta}) = -2 \log p(Y|\hat{\theta})$ is the deviance evaluated at the posterior mean $\hat{\theta} = E[\theta|Y]$.
- $p_D = E[D(\theta)] - D(\hat{\theta})$ is the effective number of parameters, representing the complexity of the model.

A lower DIC value indicates a better trade-off between fit and complexity. However, DIC is reliable primarily for models with relatively large sample sizes and asymptotic properties.

Widely Applicable Information Criterion (WAIC): WAIC is a fully Bayesian alternative to AIC and DIC, derived using the log pointwise predictive density:

$$WAIC = -2 \sum_{i=1}^{n} \left(\log E_{\theta|Y}[p(Y_i|\theta)] - V_{\theta|Y}(\log p(Y_i|\theta)) \right),$$

where:
- $E_{\theta|Y}[p(Y_i|\theta)]$ is the posterior mean of the likelihood for each observation.
- $V_{\theta|Y}(\log p(Y_i|\theta))$ is the variance of the log-likelihood over the posterior distribution.

WAIC provides a more accurate estimate of predictive performance than DIC and is particularly useful for complex hierarchical models.

- **Posterior Predictive Checks** help diagnose model misspecification by comparing simulated data with observed data.

- **DIC** is a Bayesian analogue to AIC but relies on asymptotic assumptions.

- **WAIC** is a fully Bayesian criterion that accounts for the full posterior distribution and is more reliable for complex models.

- Lower values of DIC and WAIC indicate better model fit, but they should be used in conjunction with other diagnostic tools.

These methods together provide a robust framework for evaluating Bayesian models and ensuring their predictive reliability.

6.2.3 Case Study: Bayesian Hierarchical Modeling with Real Data

We analyze real educational test score data using Python. its implementation can be found HERE

This Bayesian hierarchical approach provides more robust estimates by accounting for data structure.

Practicals

Bayesian Linear regression model

Watch this Video.
you can practice the same using the python code HERE

Bayesian Multiple Regression

Go through this Bayesian Multiple Regression
data can be found HERE

Results

Key Metrics

- $R^2 = 0.819$: The model explains 81.9% of the variance in the "Chance of Admit" variable.
- Adj. $R^2 = 0.815$: Adjusted for the number of predictors, still a strong fit.
- F-statistic = 220.9 with $p < 0.001$: The overall model is statistically significant.

Parameter Estimates

Variable	Coefficient (β)	Std. Error	t-value	p-value
Intercept	-1.2936	0.120	-10.775	<0.001
GRE Score	0.0018	0.001	3.129	0.002
TOEFL Score	0.0037	0.001	3.487	0.001
University Rating	0.0088	0.005	1.903	0.058
SOP	0.0001	0.005	0.018	0.985
LOR	0.0215	0.005	4.041	<0.001
CGPA	0.1053	0.012	8.786	<0.001
Research	0.0244	0.008	3.185	0.002

Table 6.1: OLS Regression Results

Interpretation of OLS Results

- GRE Score, TOEFL Score, LOR, CGPA, and Research experience have a significant positive effect on admission probability.
- SOP is not significant, meaning it has little predictive power.
- University Rating is marginally significant (p = 0.058).
- The residuals show some signs of non-normality (skewness = -0.955, kurtosis = 4.737), which may indicate some violations of OLS assumptions.

Variable	Mean	Std. Dev.	95% Credible Interval
Intercept	0.615	5.819	(-9.969, 10.491)
GRE Score	0.002	0.001	(0.001, 0.003)
TOEFL Score	0.004	0.001	(0.002, 0.006)
University Rating	0.007	0.010	(-0.012, 0.027)
SOP	0.001	0.005	(-0.010, 0.010)
LOR	0.022	0.005	(0.013, 0.031)
CGPA	0.104	0.012	(0.083, 0.126)
Research	0.024	0.008	(0.009, 0.039)

Table 6.2: Bayesian Regression Results

Key Bayesian Findings

- The Bayesian estimates are similar to the OLS estimates, confirming the findings.
- Bayesian credible intervals are wider than OLS confidence intervals, showing higher uncertainty quantification.
- The mean estimates align well with OLS coefficients, but uncertainty in the Intercept and University Rating is high.
- The effective sample size (ess_bulk and ess_tail) is low for some parameters, suggesting some convergence issues.

Bayesian Hierarchical Linear Regression

Go through this Bayesian Hierarchical Linear Regression
data can be found HERE

Bayesian Time series Model

Go through this Bayesian Time Series model